CRYSPY

AI BITCOIN ADVISOR

by

GROUP 4

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Introduction

This proposal covers the preliminary research for creating the front end of a cryptocurrency AI advisor, named Cryspy, for users to access on their mobile devices. This project includes the brainstorming, description, research, technical requirements, and personas developed in the process of prototyping Cryspy.

Brainstorming

(Tulsi Patel)

The AI bitcoin advisor is developed for individuals interested in investing in cryptocurrency. Before we look at the advisor application, let's look at what exactly is cryptocurrency. According to an article by NerdWallet, "Cryptocurrency (or "crypto") is a digital currency, such as Bitcoin, that is used as an alternative payment method or speculative investment. Cryptocurrencies get their name from the cryptographic techniques that let people spend them securely without the need for a central government or bank" (Rosen, A.). Now that we have a better understanding of cryptocurrency and its purpose, let's discuss how we imagine our AI bitcoin advisor application to help users navigate the crypto world.

How does the app function?

- Imagine the app working like a pop-up chat so it can be applied to different cryptocurrency mobile apps.
- The app would work similarly to how X has their AI integrated into the app, but with a Coinbase like design.
    - A separate tab at the bottom of your screen where a chat box will show, and you may ask it questions
- Security/Privacy Features: for secure transactions and protecting user data privacy
- Guide and Tutorials: to educate users on how to buy, sell and trade crypto
    - Showing how to set limits to buy or sell crypto when it reaches a set threshold
- Customer Support: for any technical problems or registering and accessing account
- Legal Updates: keeps individuals informed about the latest crypto investment laws in their residing country.
- Group Chat: where users can communicate with each other for the latest crypto news and share their insights on current investments.

What kinds of users will use this app?

- Experienced Investors: individuals who are knowledgeable in investing and the stock market.
- Inexperienced Investors: individuals with some knowledgeable of crypto investing who want to navigate the crypto market, as well as beginners with no prior knowledge or experience in crypto investments.

- Artists who auction NFTs
- Collectors who buy NFTs
- People selling actual goods and services
- Consumers buying actual goods and services
- Auction houses who help sell NFTs
- Platforms that allow the upload of NFTs, goods and services for sale/auction
- Individuals who seek to hold crypto currency short term or long term
- Users that buy decentralized land in the metaverse
- NFT holders

What sorts of questions the users might ask?

- What is cryptocurrency and how do I buy it?
- What are the risks involved in investing in cryptocurrency?
- How much funds should a new user invest in crypto?
- When is it safe to buy a certain crypto?
- What is the best strategy to earn from crypto within a shorter time frame?
- What is the difference between crypto and traditional banking?
- What are the different types of crypto?
- What is the function of a blockchain?
- What does the crypto market look like right now?
- How risky is crypto in comparison to regular stock market?
- What are the risks associated with utility tokens, meme coins, NFTs, digitalized land within the metaverse?
- What are some safe utility tokens to invest in and how long will it take to see a good amount of profit?

What sort of data would the users want to see?

- Pricing charts for cryptocurrency. YTD analysis
- When was it most often bought, and were there any specific events that influenced this buying pattern?
- Latest news on the overall crypto market or on the invested crypto
- Transaction history of buying/selling
- Risk factor of investing in a certain cryptocurrency

Description

(Christine Bonneville)

Our Cryspy AI Bitcoin Advisor will be a user-friendly pop-up chat application that can be used on mobile devices such as phones and tablets. Cryspy's intention is to provide the user with a wide variety of information to match their interest in Bitcoin. For example, it will cover topics such as the differences between the types of cryptocurrencies, how the block chain works to make this system different than traditional banking, and will be able to share what is trending in the cryptocurrency markets, along with educating the user on the risks.

Preliminary in-app features would include a glossary for terminology, crypto-market tracking, links to verified third-party educational sites, and suggestions to verified third-party online communities interested in the same type of cryptocurrency as the user, and password protection on the app itself.

Other in-app features are:

- Group chat
    - Users discussing their investments, strategies and experiences.
- Guide/Tutorial
    - Latest news of crypto market
- Graphs/data of the crypto
    - To help analyze whether a certain crypto is worth investing
- Notifications to alert on potential price drop on your invested crypto
- Customer support
- AI chatbot for questions
- live price charts
- preferred insights for crypto, NFT, and metaverse investors

Possible revenue streams for Cryspy are software offers for user from verified cryptocurrency trading platforms, where the platforms pay Cryspy to advertise their trial periods in their platforms to Cryspy's users, and tiered membership levels for novice, beginner, intermediate and advanced users.

Other revenue streams include:

- Subscription plan
    - Different benefits for each subscription plan

    - E.g., yearly plan charges less transaction fee compared to monthly plan
- For more detailed information on an interested crypto you can ask for an additional fee.
- Advertisement fee from possible crypto platforms

Future goals for My Cryspy AI Bitcoin Advisor would include eventually becoming a fully functioning platform that can be accessed on both personal computers and mobile devices for a seamless user experience. The Cryspy platform would include an education section, a community forum, a real-time ticker and graph, and a verified consumer profile where they can mint artwork, sell product, and trade in a verified cryptocurrency.

Other future goals include:

- Advanced analytics
- Expanded financial asset support
- Metaverse guidance

We believe these potential features, revenue streams, and future goals will make our Cryspy AI Advisor a promising addition to the cryptocurrency world.

Research

(Edin Husic)

Cryspy AI has four competitors but will outperform them due to their limitations.

- Cryptohopper
  - Uses an AI that analyzes the strategies you feed it and then decides which one to use. You may feed it dozens of different investing strategies and it will choose the best one. After all strategies have been imported it will rank them accordingly and choose what is best for you and the type of market you want to invest in. This also includes automatic trading – it will detect trends in the market and switch to a better investing strategy to suit the needs of the user.

    CRYPTOHOPPER

  - Screen shot of Cryptohopper Logo. (n.d.). Cryptohopper. https://www.cryptohopper.com/

- Token Metrics
  - Gives real-time market analysis as a chatbot. Can review potential coins that could explode thus profiting the user. This also offers investment guidance, but that is all it is limited to.

    TOKEN METRICS

  - Screenshot of TokenMetrics. (n.d.) TokenMetrics. https://www.tokenmetrics.com/

- InciteAI
    - AI stock and cryptocurrency trading. Uses advanced algorithms and real-time data to analyze market trends, financial metrics, and news.

    - Screenshot of InciteAI Logo. (n.d.) Inciteai. https://www.inciteai.com/

- Intellectia
    - Another chatbot where you can ask questions, and it will inform the user on market trends and give the user interactive charts rather than plain text-based bots.

    - Screenshot of Intellectia AI Logo. (n.d.). Intellectia https://intellectia.ai/

Three-quarters of competitors are chatbots/robo-advisors which may seem effective, but they do not offer the personal speculations that a human financial advisor would provide from their education and human decision-making. Once again, the chat bots are limited to what they can accomplish and Inciteai is the one advisor that is different than the other 3. Inciteai only provides AI stock analysis and nothing more yet due to regulations and/or not having the product ready for public use. Cryspy is in the beginning stage of development which raises some concern regarding risk, safety, and quality issues due to copyright violations, social offenses, and the AI hallucination phenomenon. (Morton, Koster, and Pachel, "Ask an Expert: Cutting Through the AI Hype" section, paras. 21-25).

In our plan for Cryspy, we want to keep those risks in mind as we develop it into a standalone pop-up chat bubble on a user's mobile device so the user can access it at any time regarding a cryptocurrency query.

If it does not exist, what are applications that would be of interest to a similar market?

- Hypothetically if there were no other AI crypto advisors – I think an application with human based chatbots would work, but not as well as an AI advisor. You would deal with delayed responses, wait times, limited questions could be asked,

and cannot trust another human with your crypto wallet. This could extend into AR (augmented reality) where you have virtual avatars that do what a chatbot could do, but once again it is a human on the other end of this virtual avatar.

- Voice Assistant like Siri. Eliminate the need for typing and potentially eliminate any language barriers with voice recognition and language recognition with live translation. This would revolutionize customer service, educational services, healthcare, employment services, and many more.

What might people use now to accomplish the same goals?

- Individuals could use ChatGPT, x (Grok ai), and telegram/group chats with traders. Using any of these methods could prove to be beneficial as 2 of the examples are AI agents, but they are not limited to the knowledge of crypto which could potentially hinder some results/feedback given when asked a question. The 2 AI agents are unable to invest for you due to them being designed to provide information. Telegram is a wonderful way to have dozens of individuals connecting and asking questions at the same time. Also, individuals can share their trades live with others to make profit especially with the encrypted chats. People will rely on others' information to make trades without using AI advisor agents.

Describe the features needed by your app and explain how yours would be different from existing apps?

- Features needed for our application would include
    - Basic withdraw/ deposit methods
        - Wallet ids to transfer funds across wallets
    - Smart investor
        - Cryspy will look at your preferences such as risk management, hold time and amount you're willing to invest/lose (comfortably). With this it will look at trends and up-to-date news to hopefully gain an edge in the market and allow it to make its users profit.
            - Cryspy may potentially look at upcoming legislation and give users an insight into what the market could look like based on previous events/laws.
    - Users would be able to ask any question to Cryspy that is relate to crypto
        - Legal questions for their respective geographical location
        - Bull/Bear market predictions
            - Indicators to help teach its users

        - How to distinguish the difference between a meme coin and a utility token
        - Each token on our platform would have an option to view what wallets are holding any token and how much they are profiting/losing with the associated token.
        - Users will want to see token data
            - YTD pricing
            - Insights
            - Investment risk
            - Market cap
            - Volume Traded
            - Circulating Supply
            - All time high
            - Popularity
            - AVG time users hold the coin before selling
    - Include a Cryspy coin
        - When users sign up with us and deposit, they earn a certain amount of our own Cryspy coin that they can either hold onto or swap into another token.
    - Inviting friends to the application would allow them to earn BTC/ETH
        - Incentivize for individuals to attract others

Why does it make sense to develop this app?

- It seems that all the AI crypto advisors/chat bots we researched had something in common – Each one had a feature or features the other one did not. We are here to combine all the features into one application before any other to try and capture as many users as possible. Users tend to like one application and not have to switch between multiple applications because one has a feature that the other does not. Some applications/AI advisors may be close to our design, but we are dedicated to being user friendly and including everything a novice or expert investor requires.
- Chat bots are also limited to the amount of information they can give and limited to what they can do for the user. It's a no brainer to develop this app and provide users with a real AI crypto advisor rather than a chatbot that may potentially mislead its users to earn profits.

Technical Requirements

(Micheal Owuso-Adiefe)

The Cryspy AI bitcoin advisor will be a mobile application with a user-friendly interface that can be used on tablets and phones. The user will use touch and swipe inputs to easily navigate the app. Users will also be able to customize the number of push notifications they receive from the app.

Hardware Features:

- **Touchscreen:** The app will rely on touch and swipe inputs for navigation.

Device Constraints/Advantages:

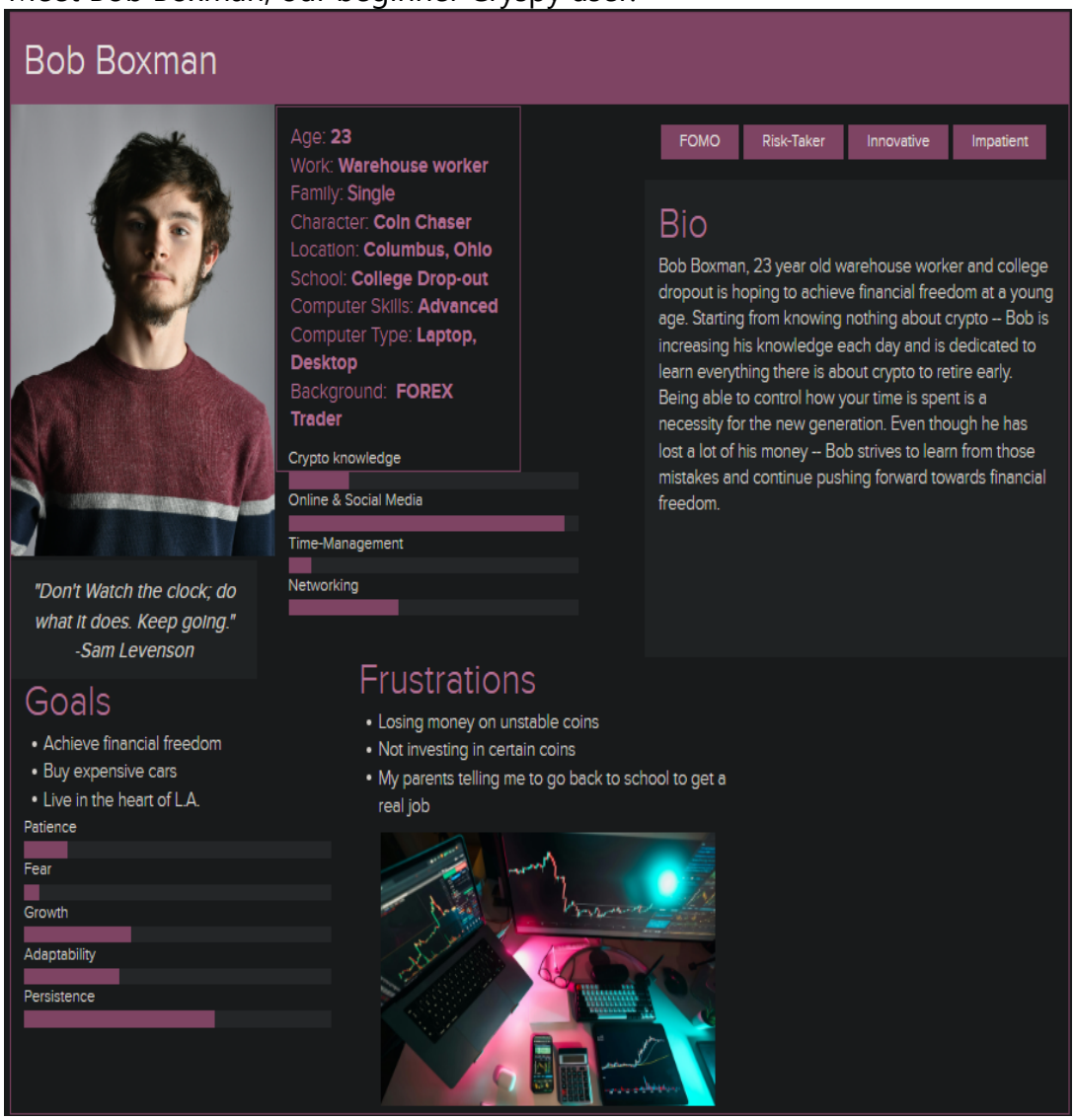
- **Advantages:**
  - Mobile devices are portable, users can access crypto data on the go
  - Push notifications can be used to alert users about significant price changes, news, or updates.
  - Encourage user engagement
  - Convenience

- **Constraints:**
  - Limited screen size may make it hard to display data or complex charts.
  - Battery consumption could be an issue
  - The app may cause heavy mobile data usage
  - May consume a large amount of storage space

Personas

For our three personas, we have one beginner, one intermediate, and one advanced.

Meet Bob Boxman, our beginner Cryspy user:



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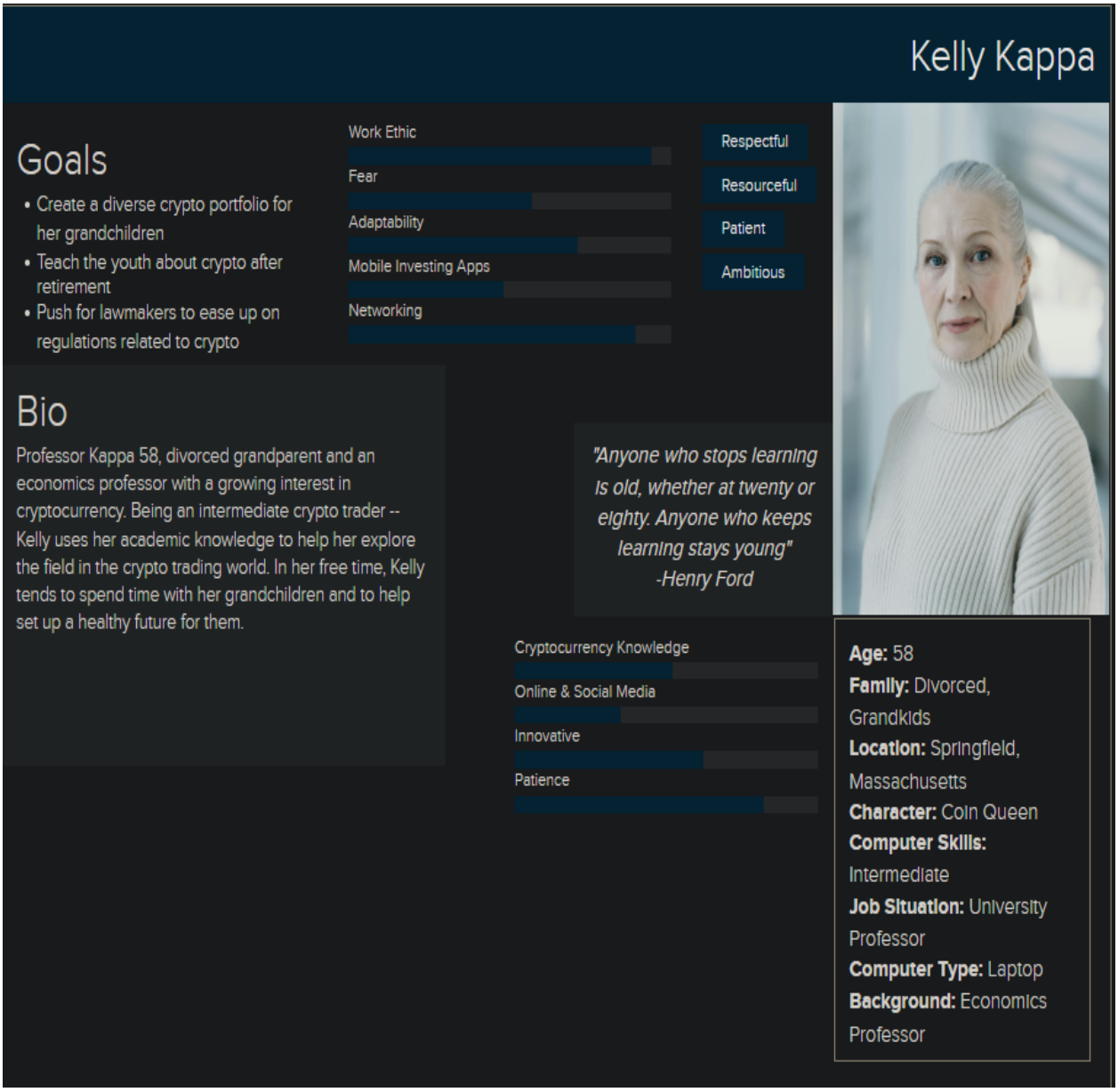
Meet Kelly Kappa, our intermediate Cryspy user:



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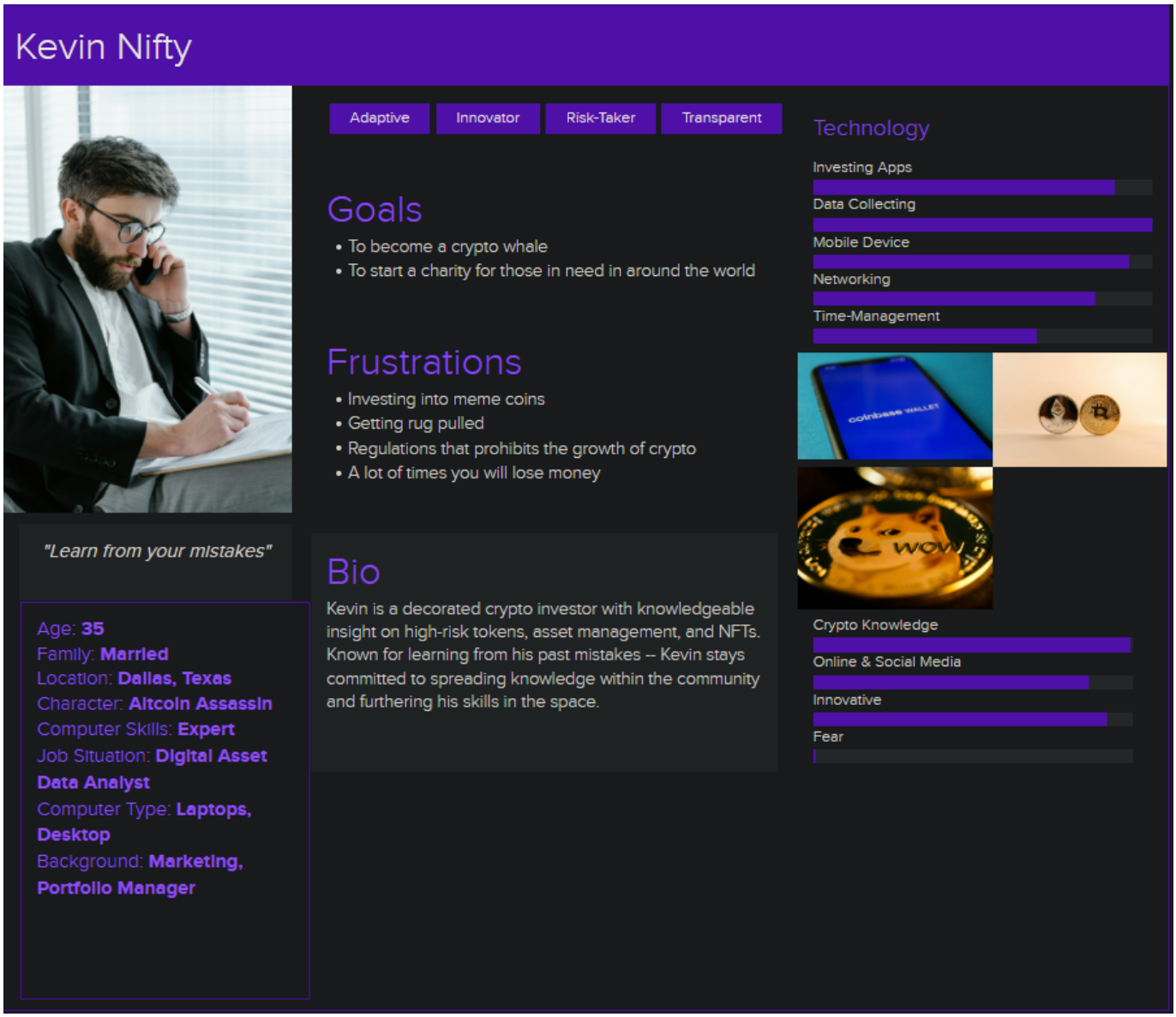
Meet Kevin Nifty, our advanced user:



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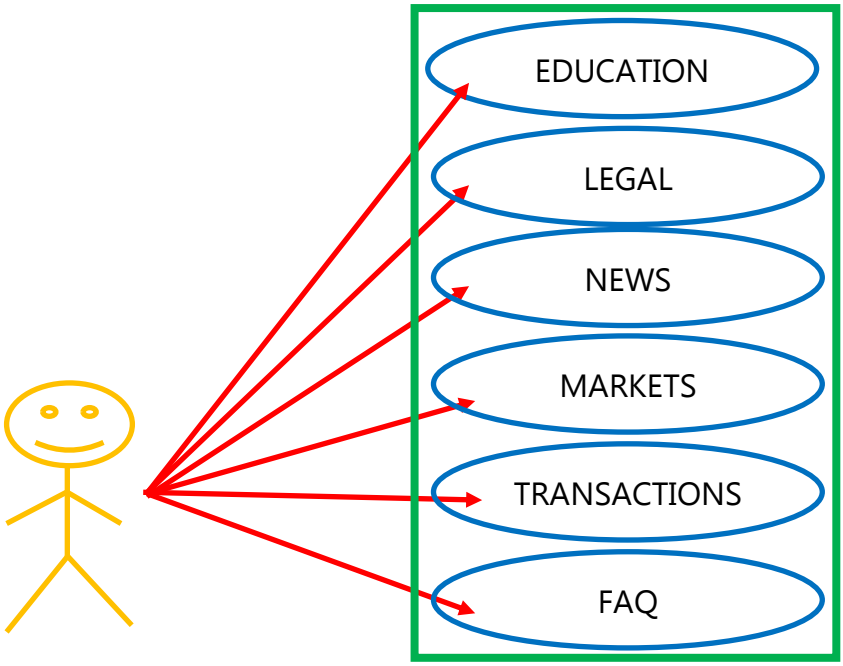
Use Cases

There were fifteen use cases that were some good possibilities for the functions of the Cryspy AI Bitcoin Advisor:

1. inquire about how crypto and blockchain works
2. how to diversify their portfolio
3. show the performance trends for daily, weekly, monthly
4. artist who inquiries about minting their artwork (guides/tutorial)
5. small business owners who want to inquire about listing product (guide/tutorial)
6. latest news updates
7. legal updates for specific areas
8. transaction history
9. risk management FAQ section
10. customer support link in case Cryspy doesn't know
11. group chat/forum link
12. inquiries about the types of investment options (utility tokens, meme coins, NFTs, digitalized land in metaverse
13. difference between traditional and crypto banking/investments
14. difference between stock and crypto markets
15. teaching the crypto market trends

To simplify the use cases, they were grouped by broader topic that the user would be interested in.



Use Case 1

Use Case Name	check the crypto latest news	
Brief Description	Cryspy will inform the user about the latest news on crypto, depending on the user's needs.	
Actors	Kevin Nifty	
Normal Flow of Events	Actor	System
	1. Open the app	2. Will ask what you want to see today?
	3. Check the latest news	4. Which type of crypto news are you interested in?
	5. Give me legal news on the big three coins, BTC, Eth, Sol	6. Are you looking for current or future legal news?
	7. I would like to see present news for crypto	8. Redirects to a news feed and Gives headlines with links on legal news about the big 3 coins
		9. Is there anything else I can help you with?
	10. No	11. Great, have a good day!

Use Case 2

Use Case Name	Education on the big three coins, BTC, Eth, Sol	
Brief Description		
Actors	Bob Boxman	
Normal Flow of Events	Actor	System
	1. What are the three big coins?	2. BTC, Eth, Sol. Would you like more info on them all or just one?
	3. Info on all of them	4. Cryspy gives pricing for the tokens.
		5. Would like you to know about the market cap for these coins?
	6. Yes	7. Gives info on market cap.
		8. Would you like to know about the volume being traded?
	9. Yes	10. Give info on volume being traded
		11. Would you like info on daily, weekly and monthly performance of these coins?
	10. yes	12. Give info on performance for week, day, month.
		13. Will that be everything for today?
	14. Yes	15. Great, have a good day!

Entities

These are the tables that describe the entities involved in our use cases:

Entity: AI Agent	
Name	Cryspy
Version	1.0
APIs (Application programming interface)	User ID, search, pop-up chat, login, app appearance settings, user crypto preferences, market data,

Entity: Tutorials	
Title	Welcome to Cryspy AI Advisor
Duration	User determined
Category	Introductory
Format	Slides for the APIs
Difficulty	Easy

Entity: Conversation Thread	
User ID	Set by user in API for User ID
Topic	Education on the big 3 coins
Date Created	February 1, 2025
Conversation ID	CTEM54321
Status	open or closed

Entity: Portfolio of Crypto Product	
Portfolio/Wallet ID	E2T6M8C9
Risk Level	Medium
Profit/Loss	+
Total Value	$2,000
Performance	Moderate + 30%

Content Plans

This the general layout for each page in the app:

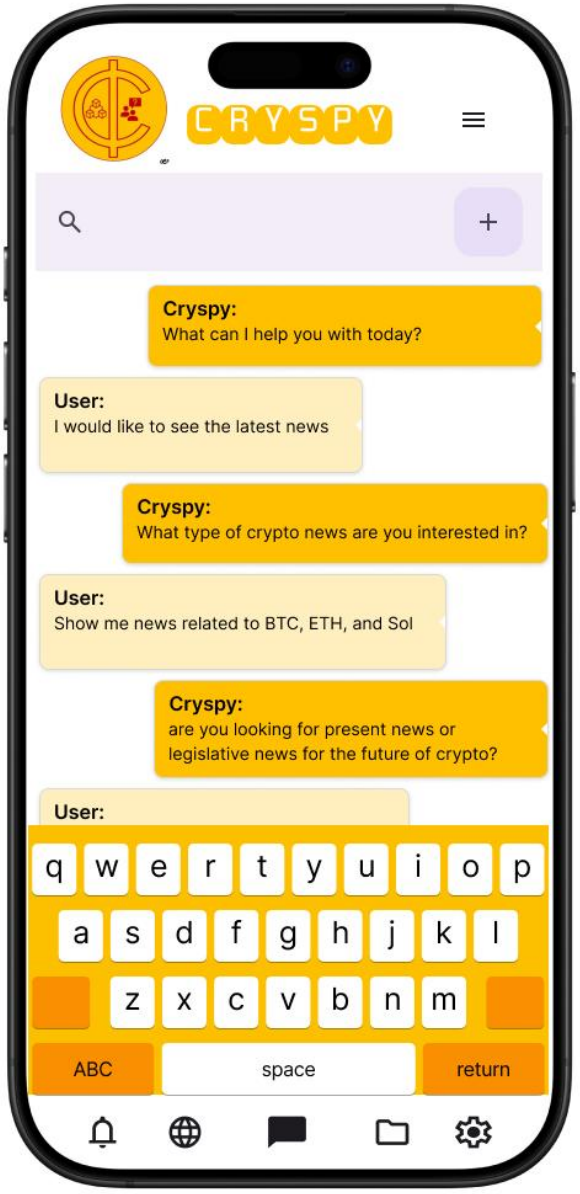
Cryspy Pop-Up Chat App

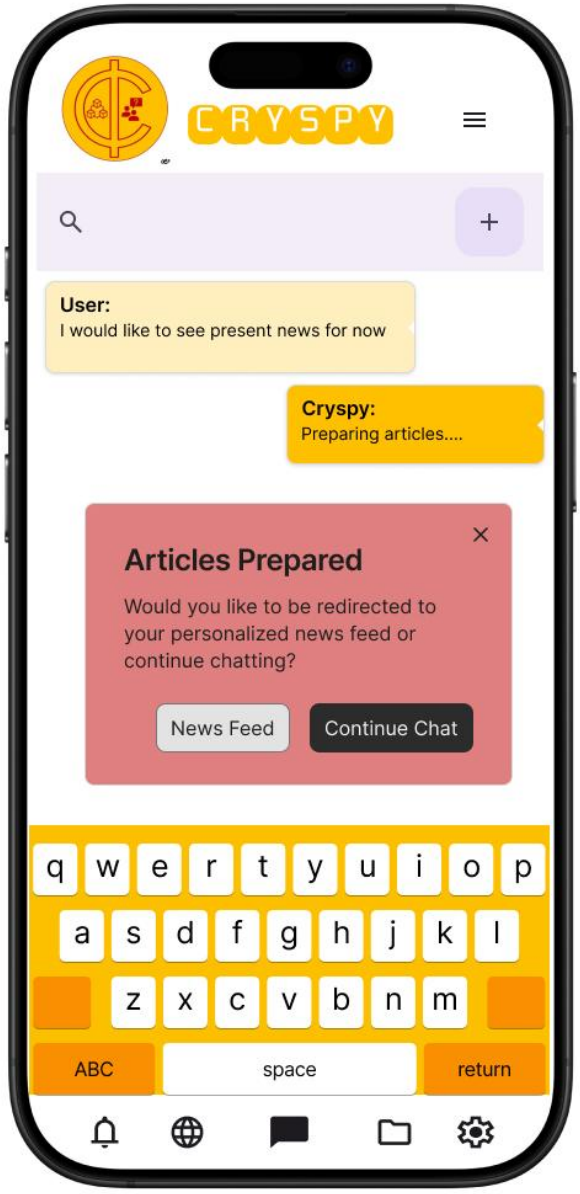
1. Header
2. Hamburger
    a. Sign In
    b. Account settings
        i. User crypto preferences
        ii. App appearance preferences
        iii. Accessibility
        iv. Security & Privacy
        v. New To Cryspy Tutorial
        vi. Cryspy Chat History
3. "Ask Cryspy" search bar
4. Main Content Panel
5. Icon Quick Links
    a. Notifications
    b. News
    c. Chat history
    d. Saved Transactions
    e. Settings
        i. Customer Support
        ii. Contact us
        iii. Troubleshooting
        iv. Report an Issue

Prototypes

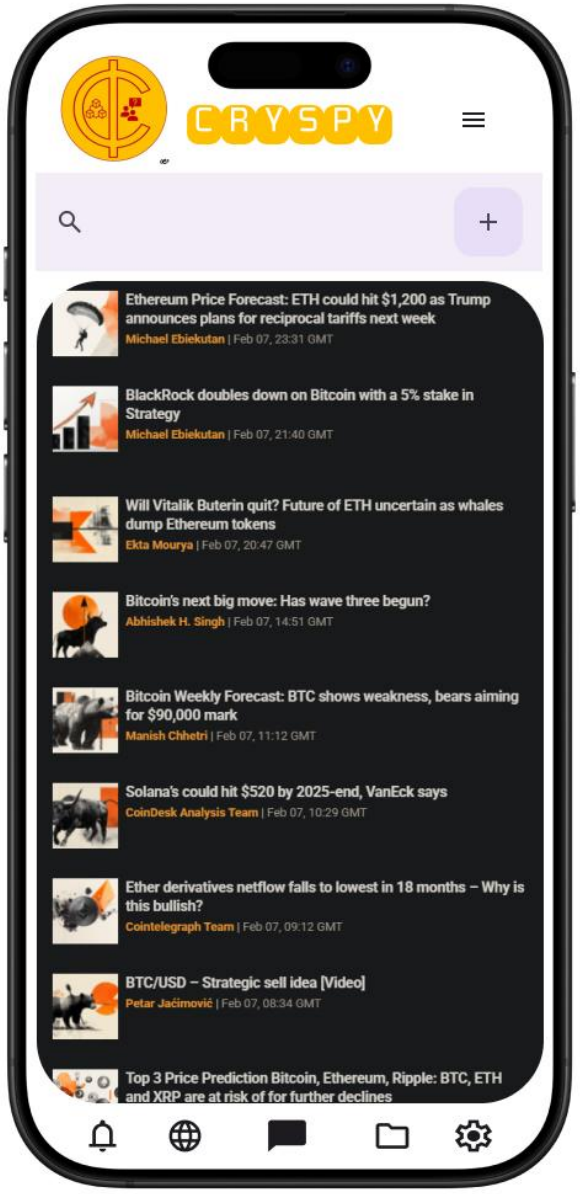
These are the Figma prototypes for the Use Case 1, where Kevin Nifty wants to get information on cryptocurrency news, and Use Case 2, where Bob Boxman is asking question about the values about the three big coins.

Use Case 1, Frames 1 and 2:





Use Case 1, Frames 3 and 4:



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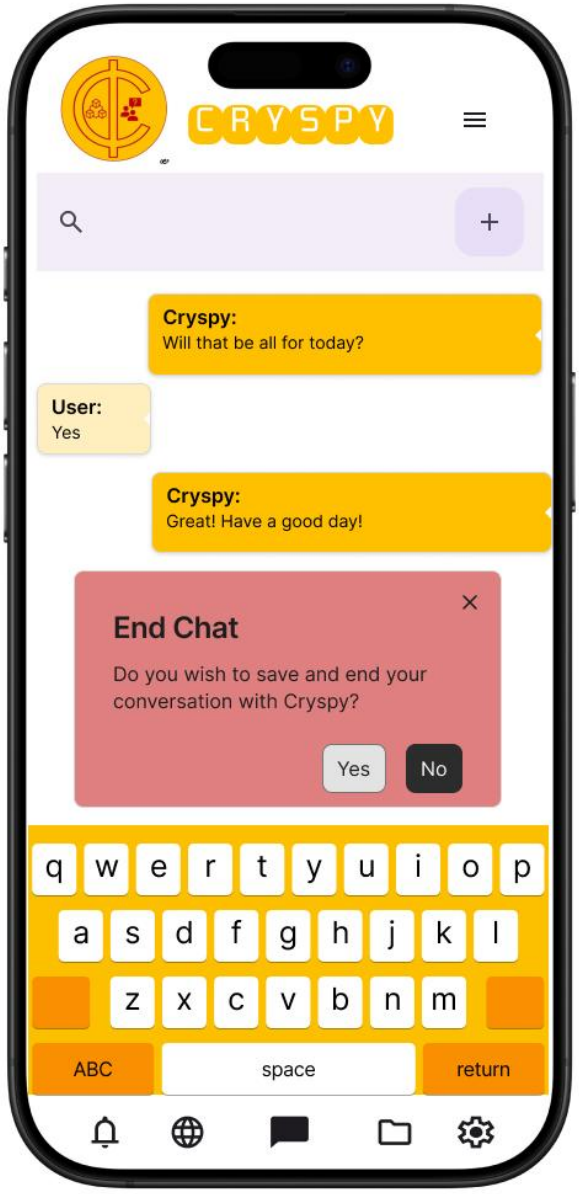
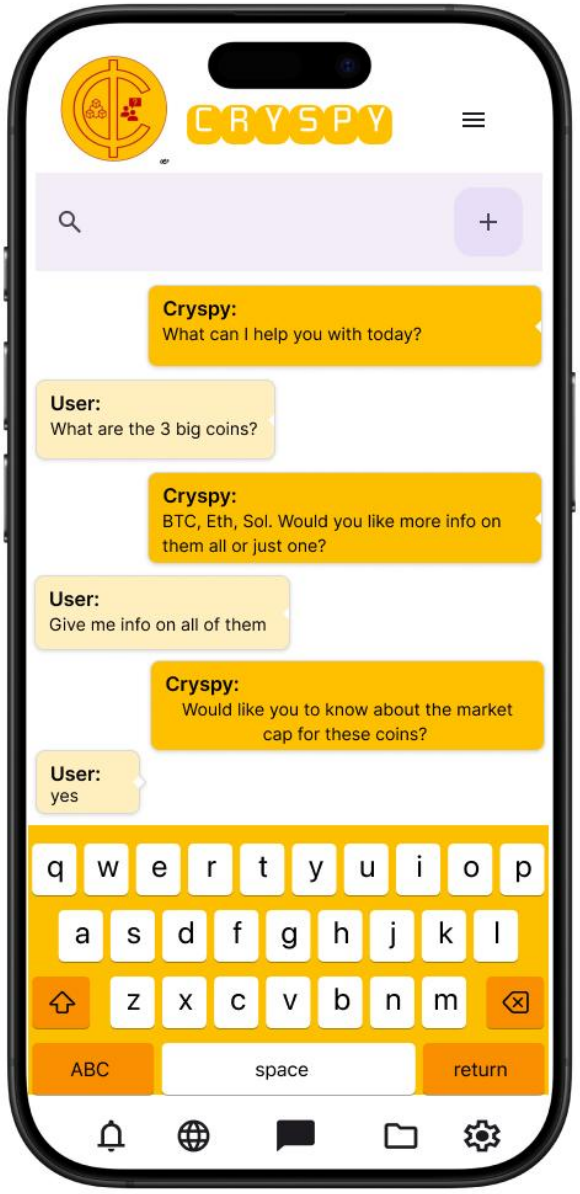
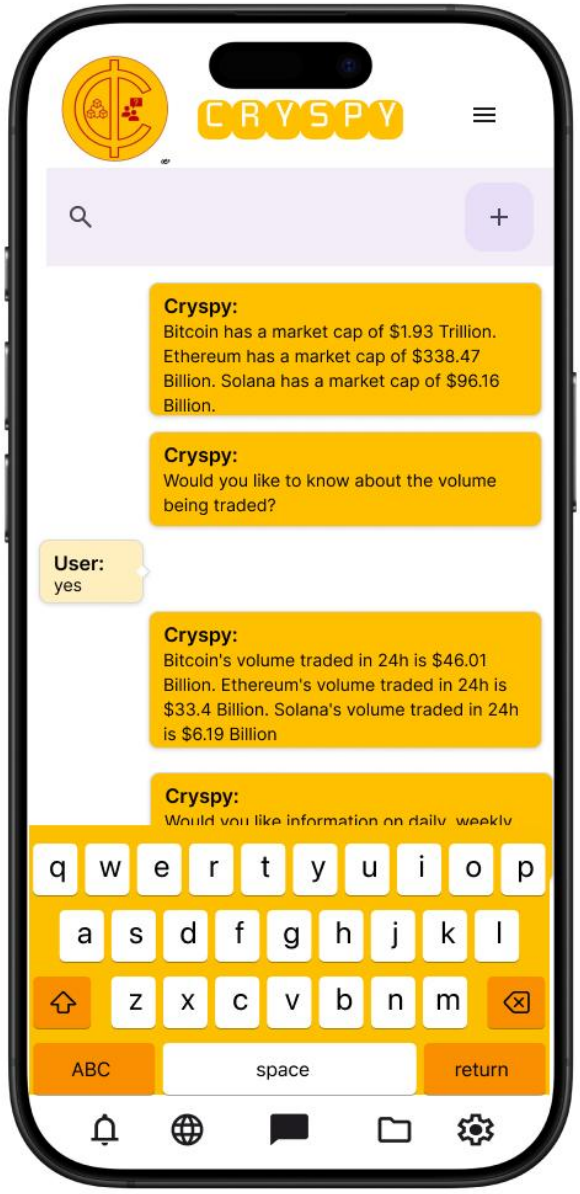
Use Case 1, Frame 5:

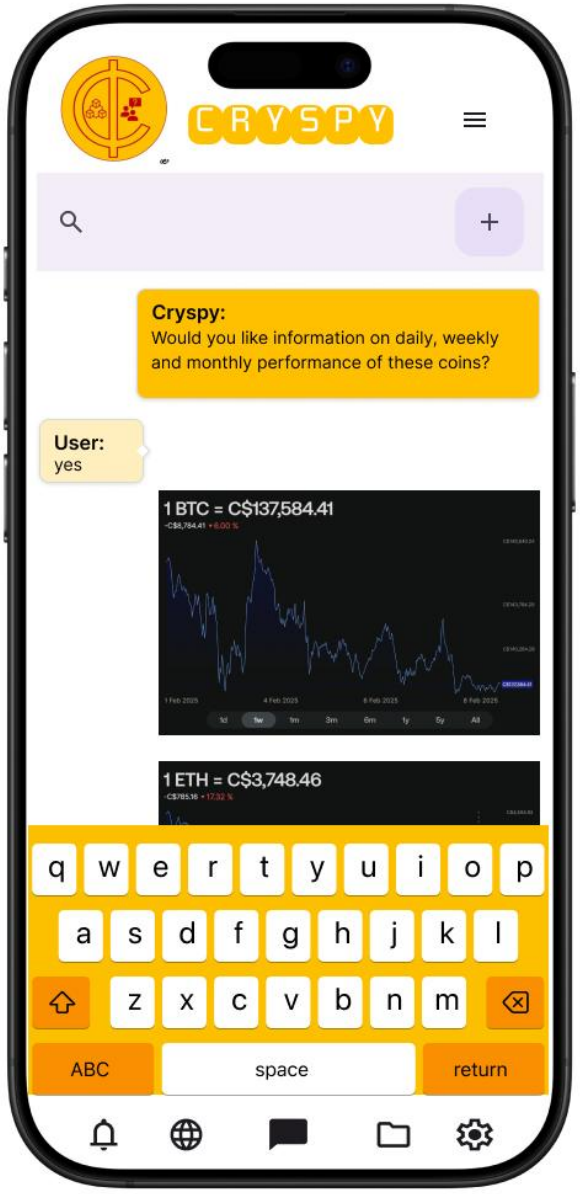


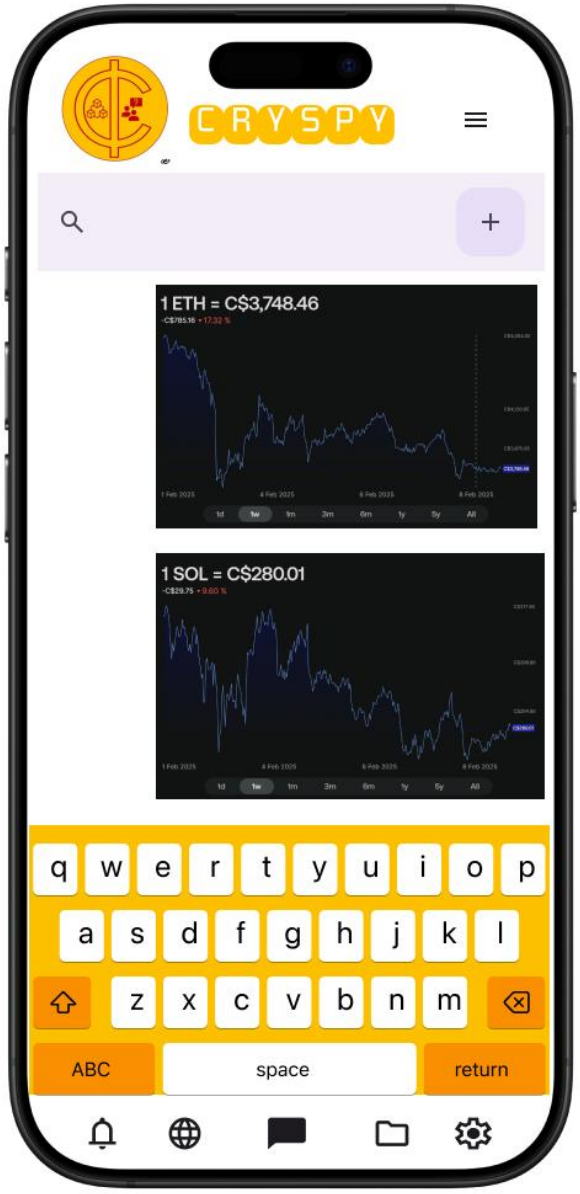
Use Case 2, Frames 1 and 2:





Use Case 2: Frames 3 and 4:





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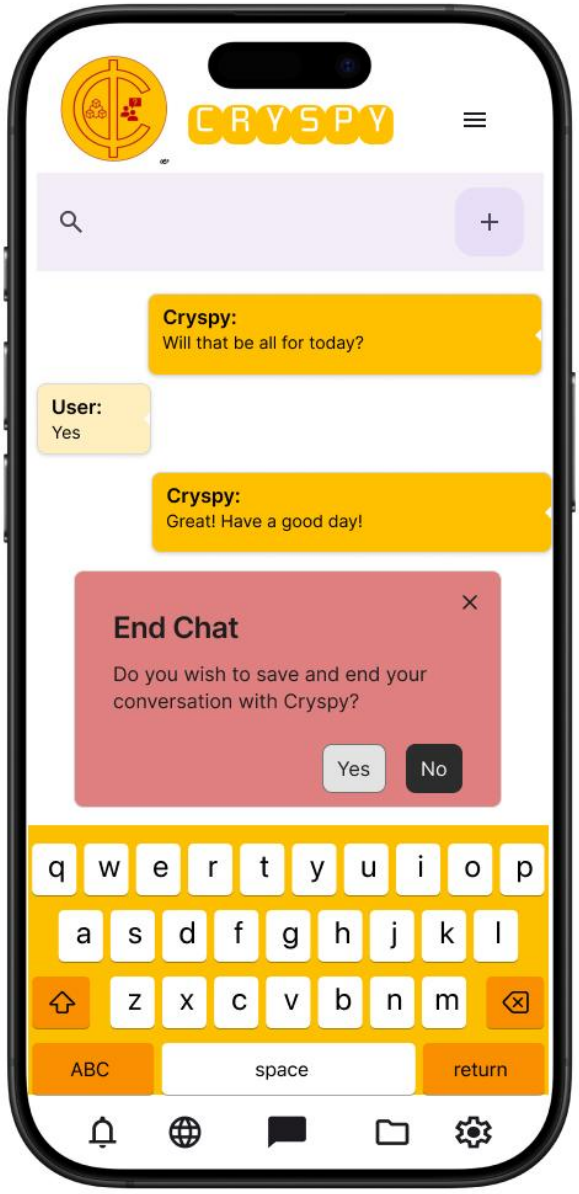
Use Case 2, Frame 5:



Conclusion

In conclusion, we feel that our Cryspy AI Advisor with its educational and user-friendly features will be an asset to the exciting world of cryptocurrency, more accessible to a broad spectrum of users through their personal mobile devices.

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